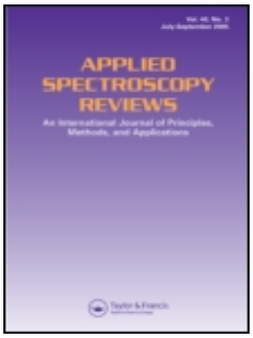

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Amphiphilic metabolites in gallbladder bile: Potential biomarkers for gallbladder diseases

Hyeji Ko[a], Ikjang Choi[a], Kyeol Chang[b], Gijin Jeong[a], Gyeonghyeon Gong[a], Hyeonglim Seo[a], Donghyun Ryu[b], Kyeong Geun Lee[c], Dongho Choi[c], Hoeil Chung[b], and Youngbok Lee[a]

[a]Department of Applied Chemistry, Hanyang University, Ansan, South Korea; [b]Department of Chemistry, Hanyang University, Seoul, South Korea; [c]Department of Surgery, Hanyang University Hospital, Seoul, South Korea

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ABSTRACT
Gallbladder bile is one of the most abundant body fluids, and metabolic compositions of the bile are highly correlated with several gallbladder diseases (gallstones, gallbladder polyps, cholecystitis, and biliary tract cancer). The gallbladder diseases are generally diagnosed by several different imaging methods in the clinic; however, none of them can readily reveal detailed information about the diseases in molecular levels. Here, we have applied various nuclear magnetic resonance spectroscopy in order to identify and analyze composition of the human gallbladder bile, since the spectroscopic method provides not only structural information but also dynamic information of low- and high-weighted metabolites. In combination with both 1D Carr–Purcell–Meibom–Gill filtered ^{1}H spectrum and 2D ^{1}H–^{13}C heteronuclear single quantum correlation spectrum, 15 metabolic compounds have been assigned in the bile specimen. Discrimination and classification analysis have been conducted by principal component analysis and support vector machine, respectively, so as to differentiate the gallbladder diseases, especially between gallstones and gallbladder polyps in here. From these investigations, we found two family of metabolites, namely bile acids (glycine and taurine conjugated cholic acids) and phosphatidylcholine, which play significant roles in discriminating gallstones, gallbladder polyps, and others.

KEYWORDS
NMR spectroscopy; discrimination analysis; gallbladder disease; metabolic profiling

Introduction

Gallbladder bile is the most abundant bio-fluids, which is produced at liver and stored in gallbladder. Main compositions of the bile are water (82%), bile acids (12%), phospholipids (4%), un-esterified cholesterol (0.7%), and others (1, 2). Several bile components, especially bile acids and phospholipids, play a pivotal role in managing many water insoluble metabolites (fat, cholesterol, and others) (3). The surfactant-like molecules prevent the fat-like metabolites to be crystallized by forming micelles or vesicles between them (4–7). Therefore, the constituents and concentrations of bile are strongly controlled or maintained in

CONTACT Youngbok Lee yblee@hanyang.ac.kr Department of Applied Chemistry, Hanyang University, 55 Hanyangdaehak-ro, Sangnok-gu, Ansan, Gyeonggi-do 426-791, South Korea.

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healthiness, however, malfunction of the manipulation may cause several bile-related diseases such as, gallbladder cancer, gallstones, cholecystitis, and gallbladder polyps (8,9). For example, recent researches suggested that the two cases (*either* accumulation of cholesterols or bilirubin, *or* deprivation of bile acids or phospholipids) induces crystallization process of fats or cholesterols, inducing gallstones (10–12). In addition, gallstones may gradually induce cholecystitis with a bacterial infection, conclusively resulting in gallbladder polyps (benign or malignant) (13). In these regards, further investigations of the bile components in molecular levels are quite needed to acquire valuable diagnostic tools, which can contribute to informing specific pathological conditions.

In clinic, the bile-related diseases can routinely be diagnosed by several different imaging techniques such as ultrasound, computed tomography, magnetic resonance imaging (14–16). However, the common imaging techniques are not so effective in some cases, for example, it is hard to distinguish the gallstones from the surrounding tissues when the gallstones have a soft character (17). In addition, only few limited information about metabolites in molecular level representing pathological conditions would be obtained from the imaging method (18).

From this point of view, new diagnostic tools with high sensitivity and selectivity are indeed desired to provide detailed information related with the gallbladder diseases in molecular levels. Several spectroscopic techniques in combination with metabolomics and/or proteomics approaches have been developed for the purposes of biomaterial examination (19–24). For instance, mass-based analysis offers an excellent sensitivity and selectivity platform of proteomics research for detailed diagnosis of the diseases, nevertheless, the spectroscopic method strongly rely on performances of separation or ionization steps so as to obtain robust quantitative investigation, namely requiring extensive optimizations and pretreatments processes (25–27). On the other hand, high-resolution nuclear magnetic resonance (NMR) spectroscopy has long been played a significant role in the study of bio-fluids (urine, blood, bile, and others) because the technique reveals detailed structural and dynamical information of small metabolites with atomic resolution (28–30). The most prominent advantages of the NMR spectroscopy are its useful characteristics for analysis in quantitative manners and with minimal or no sample preparation. The properties of metabolites are thoughtfully reflected in the chemical shifts (^{1}H and ^{13}C; typically 10 ppm and 200 ppm ranges, respectively), which are dependent on the electronic structure around each nucleus (shielding effects). Moreover, several MNR parameters such as, T_1/T_2 relaxation times, *J*-coupling, and nuclear overhauser effect (NOE), and others, are very valuable as probes to investigate the molecular information (31–33). In spite of the strengths, the practical usage is sometimes restrained by its low sensitivity (need micro-molar concentration at least) or spectral overlapping, requiring the use of high-field NMR instruments equipped with a cryogenically cooled probe (34).

Here, we have conducted a feasibility test on NMR spectroscopy-based differentiation of gallbladder diseases using powered gallbladder bile samples, focusing on metabolic identification and differentiation, especially for between gallstones and gallbladder polyps. One-dimensional Carr–Purcell–Meibom–Gill (CPMG) filtered ^{1}H spectra and two-dimensional ^{1}H–^{13}C heteronuclear multi-dimensional NMR spectra were analyzed for metabolic profiling of the gallbladder bile specimen on a common instrumental setup (400 MHz NMR instrument equipped with a broadband probe). Furthermore, we have performed more detailed discrimination analysis (PCA) in combination with the metabolic search, pointing out foremost biomarkers which provide a significant role in discrimination gallstones, gallbladder polyps, and others.

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Experimental

Patients

We obtained all of gallbladder bile samples from patients with gallstone, gallbladder polyp, and others at Hanyang University Hospital (Seoul, Korea). The sample sets consisted of specimen from 21 gallstone (11 males and 10 females, age ranging from 31 to 77 years), seven gallbladder polyp (four males and three females, age ranging from 24 to 63 years), two cholecystitis (two females, age 23 and 77 years), and four normal (three males and one female, ages ranging from 40 to 65 years). There were no any correlations between the patient groups corresponding to gender and age. The clinical data of the patients and samples were collected properly after the permission from institutional review board of Hanyang University.

Sample preparation

The bile sample was taken from the gallbladder after cholecystectomies, and frozen by liquid nitrogen and stored at −80°C. For the NMR analysis, the intact bile sample was thawed and lyophilized. Before the freeze–drying process, insoluble components were clearly removed in advance by a centrifugation (10,000 rpm for 30 min at 4°C). 10 mg of the resulting powder was dissolved in 500 μL of potassium phosphate buffer solution (PBS, pH 7.3, buffer strength 100 mM, containing 50 μL of D_2O). The aliquot was transferred into a 5 mm NMR tube (o.d.), and the sample was stored at 4°C until further analyzed. Total processes of the sample preparation in the buffer solution were summarized in Figure 1. In order to compare spectral features between aqueous and organic media, 4 mg of the powder was also prepared in 500 μL of deuterated methanol (CD_3OD). Since solubility of the bile components was fairly low in the organic solvent, insoluble components were separated by a centrifugation (2,000 rpm for 30 min at 4°C) before NMR measurements.

NMR spectroscopy

All of the 1D ^{1}H spectra were measured by single pulse sequence with a solvent pre-saturation (water: 4.7 ppm) on a Bruker 400 MHz NMR spectrometer equipped with a broadband probe containing three pulsed field gradients (Bruker Biospin, Billerica, MA) at a temperature of 27°C. In order to selectively remove broad background signals originated from

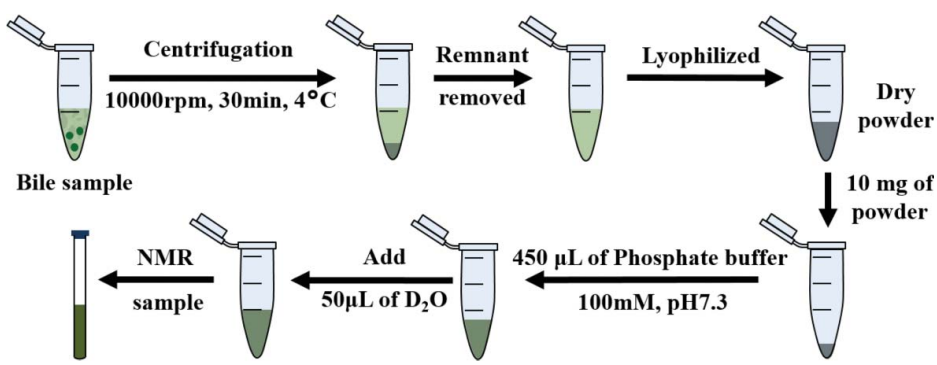

Figure 1. Processes of sample preparation of gallbladder bile for NMR spectroscopy.

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high-weighted metabolites, 100 ms of CPMG filter was applied before the single $\pi/2$ excitation pulse (the pulse strength was 22.52 kHz). For each scan, 32,892 data points were obtained for an acquisition time of 2.72s. Solvent suppression was achieved by pre-saturation, recycle delay in between each scan set to 10s. The raw data were zero filled to 65,536 complex data points, and an exponential window function with 0.30 Hz line broadening was applied before Fourier transform using TOPSPIN 3.5 program (Bruker Biospin, Billerica, MA). Two-dimensional multiplicity-edited ^{1}H–^{13}C heteronuclear single quantum correlation (HSQC) spectrum was obtained by the pulse with spectral editing technique (35,36). For the transverse relaxation (T_2) experiments, a Carr–Purcell–Meibom–Gill (CPMG) pulse sequence was used, and the relaxation time was calculated from fit to a single exponential function (37,38). Chemical shifts were calibrated with an internal standard (trimethylsilyl propanoic acid (TSP) signal at 0 ppm).

Discrimination analysis

PCA was performed using MestReNova 10.0 (Mestrelab Research, USA) (39,40). Before conduct the discrimination analysis, all of the raw NMR spectra were pre-processed with binning, aligning, and normalization functions. Discrimination accuracy was determined by support vector machine (SVM) using MATLAB R2014a (MathWorks Inc., MA, USA) (41). Gaussian kernel was used in SVM, and the corresponding bandwidth (h) of Gaussian kernel was 0.001 and the penalty parameter(C) was 1 for this study.

Results and discussion

NMR spectral features of gallbladder bile

We obtained conventional 1D ^{1}H spectra of gallbladder bile samples dissolved in buffer by using single-pulse NMR experiment. Solvent suppression (δ ^{1}H 4.7 ppm) was conducted with a water gate pulse sequence. For spectral comparisons, two spectra from gallstone and gallbladder polyp groups were only selected and displayed in Figure 2a. Numerous broad signals were detected in ranging from 4.5 ppm to 0 ppm, which were mainly originated from aliphatic signals of various metabolites. Noticeable spectral highlights between the two groups were not able to be directly assigned due to the broad spectral lines, however, the spectral features were fairly different over the entire spectral range. In particular, signal intensities at both 0.5–1.5 ppm and 3.2–3.3 ppm were moderately different even though the difference was not able to be quantitatively addressed due to the signal overlapping. The broad spectral features are commonly observed in ^{1}H NMR spectrum when macromolecules such as, protein, polymer, or micelle, are presence in the sample. Since the main components of the gallbladder bile are water, bile salts, phospholipids, fats, bilirubin, and inorganic salts, it is expected that the broad spectral patterns were stemmed from an aggregation of metabolites between amphiphilic surfactants (bile salts and phospholipids) and oil-type molecules (fats and cholesterols), namely a form of micelle. So as to confirm presence of the micelles in buffer system, we prepared the same bile sample dissolving in deuterated methanol, and compared spectral shapes between of them in terms of linewidths (Figure 2b). As indicated in Figure 2, the bile sample in the aqueous solution showed much broader linewidths than the sample in deuterated methanol as a result of existence of the micelles. Several sharp

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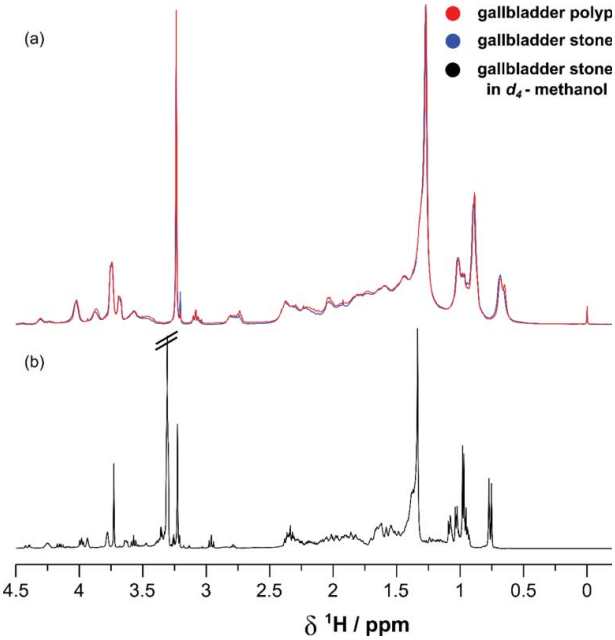

Figure 2. One-dimensional ^{1}H NMR spectra of (a) gallbladder polyp (red) and gallstone (blue) in buffer solution. (b) One-dimensional ^{1}H NMR spectrum of gallstone (black) in deuterated methanol.

signals of low-mass metabolites could be obviously observed in deuterated methanol, however, it should be pointed out that signal intensities in the organic solvent are not able to represent any quantitative information about the metabolites due to their low solubility. In addition to the simple linewidth comparisons, we examined a more detailed analysis, specifically transverse relaxation time (T_2) analysis, which is a valuable NMR parameter indicating local mobility or flexibility of molecular system investigated. Since the molecular motion and mobility of the micelle are very limited, very fast T_2 relaxation times from the sample in buffer system (around few millisecond ranges, and cannot be precisely determined in some cases) were observed over the entire spectral regions, confirming presence of the micelles. In contrast, the identical sample in the organic solvent represented relatively slow T_2 relaxation times (overall 600 milliseconds of T_2 relaxation time) owing to fast molecular tumbling time and unconstrained molecular motions, indicating metabolites in non-aggregated forms.

Metabolite assignments in the gallbladder bile

The aggregation of bile components in micelles and their severe spectral overlapping hinder precise detection and identification of the bile components. In order to resolve the issue, a CPMG filtered experiment was applied since broad background-like signals can be selectively decreased due to their shorter T_2 relaxation times. As specified in Figure 3a, chemical shift information about low-weighted metabolites could be clearly observed in the CPMG spin-echo spectrum, and their presences were identified by both Chenomx 8.1 NMR Suite software (Chenomx, Inc. Edmonton, Canada) and metabolite libraries (Figure 3b and Table 1) (42,43). However, the use of 1D ^{1}H NMR spectroscopy is exceedingly limited for assignments for high-weighted metabolites like bile acids or phospholipids, due to their

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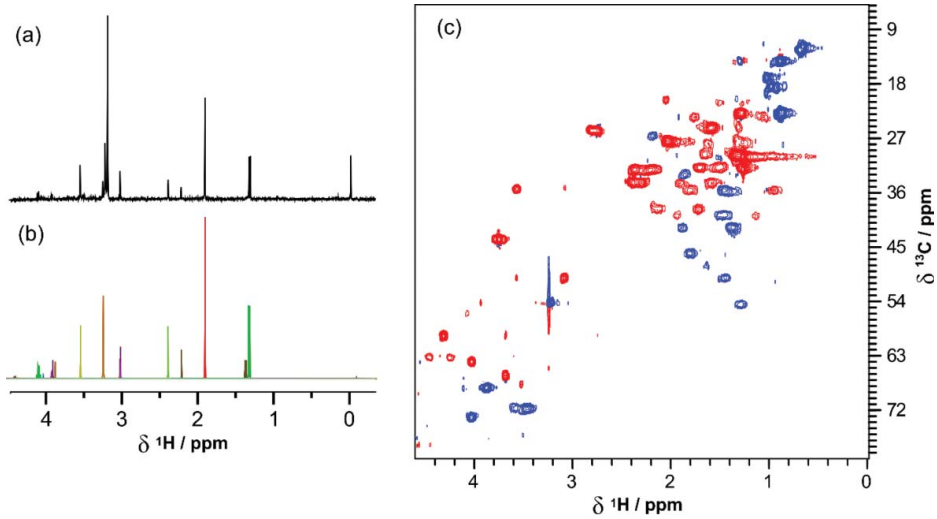

Figure 3. (a) One-dimensional CPMG filtered ^{1}H NMR spectrum of gallstone sample. (b) Simulated ^{1}H spectrum of small metabolites which were assigned in (a). (c) Two-dimensional ^{1}H–^{13}C HSQC NMR spectrum of the gallstone sample. For clear peak assignments, phase sensitive module was implemented in the pulse sequence (CH_3, CH groups are represented with positive signals (blue color), and CH_2 groups are represented with negative signals (red color)).

spectral overlaps, making frequent use of multi-dimensional NMR spectroscopy necessary. For the purpose of detailed metabolic profiling of the metabolites, two-dimensional multiplicity-edited ^{1}H–^{13}C HSQC spectrum was obtained, where CH/CH_3, and CH_2 spin systems were represented with positive, and negative signal intensities, respectively (Figure 3c). In the two-dimensional spectrum, major bile acids (cholic, deoxycholic, and chenodeoxycholic acids) conjugated with glycine or taurine were distinctly detected, and their chemical shits

Table 1. Chemical shift information of assigned low-weighted metabolites in gallbladder bile.

Name	δ ^{1}H/ppm	δ ^{13}C/ppm	Assignment
Acetate	1.92	23.5	CH_3
Betaine	3.27		$N–(CH_3)_3$
	3.90		CH_2
Carnitine	3.23		$N–(CH_3)_3$
Choline	3.20	54.1	$N–(CH_3)_3$
	3.51		CH_2
	4.03	64.0	CH_2
Creatine	3.05		$N–CH_3$
	3.93	54.4	$N–CH_2$
Creatine phosphate	3.04		$N–CH_3$
	3.95		$N–CH_2$
Creatinine	3.05		$N–CH_3$
	4.05	56.1	$N–CH_2$
Formate	8.46		CH
Lactate	1.33	20.6	CH_3
	4.10	68.4	CH
Leucine	0.82	22.9	CH_3
Succinate	2.45	34.2	CH_2
Valine	0.97		CH_3
	1.02	19.7	CH_3

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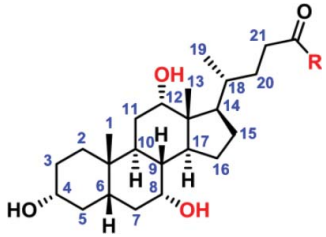
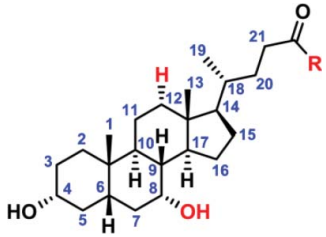
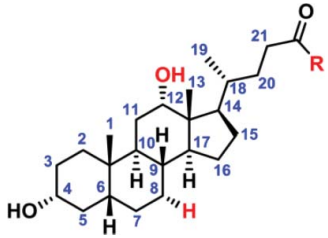
Table 2. Chemical shift information of various cholic acids in gallbladder bile.

Name	Cholic acid		Chenodeoxycholic acid conjugates		Deoxycholic acid conjugates	
Structure						
Assignment no.	δ ^{1}H/ppm	δ ^{13}C/ppm	δ ^{1}H/ppm	δ ^{13}C/ppm	δ ^{13}C/ppm	δ ^{1}H/ppm
1	0.89	22.8	0.89	22.8	0.89	22.8
2	0.95	35.7	0.95	35.7	0.95	35.7
3	1.27, 1.65	27.7, 29.6	1.27, 1.65	27.7, 29.6	1.27, 1.65	27.7, 29.6
4	3.46	71.7	3.46	71.7	3.60	71.5
5	2.13, 1.70	38.6, 41.4	2.13, 1.70	38.6, 41.4	1.81	35.5
6	1.37	41.7	1.37	41.7	1.30	42.2
7	1.60, 1.91	34.4	1.60, 1.91	34.4	1.27, 1.89	27.6
8	3.88	68.2	3.88	68.2	1.32	25.1
9	1.47	39.7	1.40	39.7	1.46	35.6
10	2.18	26.6	1.85	34.3	1.85	34.3
11	1.63	27.7	1.22, 1.52	21.0	1.58	28.2
12	4.02	73.1	1.94, 1.14	39.8, 39.9	4.02	73.1
13	0.69	12.4	0.66	11.9	0.69	12.4
14	1.80	45.8	1.29	54.5	1.80	45.8
15	1.27, 2.03	27.6, 27.5	1.27, 2.03	27.6, 27.5	1.27, 2.03	27.6, 27.5
16	1.08, 1.76	23.3, 23.5	1.03, 1.71	23.8, 23.7	1.03, 1.71	23.8, 23.7
17	1.88	41.8	1.44	50.0	1.63	47.8
18	1.46	35.6	1.46	35.6	1.46	35.6
19	1.02	17.2	0.98	18.3	1.02	17.2
20	1.51, 1.70	31.9	1.51, 1.70	31.9	1.51, 1.70	31.9
21	2.20, 2.36	32.1, 32.2	2.20, 2.36	32.1, 32.2	2.20, 2.36	32.1, 32.2
R			N–CH_2 conjugated taurine (δ ^{1}H/ppm, δ ^{13}C/ppm): 3.57, 35.3 S–CH_2 conjugated taurine (δ ^{1}H/ppm, δ ^{13}C/ppm): 3.09, 50.0 CH_2 conjugated glycine (δ ^{1}H/ppm, δ ^{13}C/ppm): 3.75, 43.7			

R = glycine or taurine.

were listed in Table 2. However, it was not evidently distinguishable between glycine and taurine conjugations on a specific bile acid because the conjugation have not great effects on chemical shifts change on the bile acid moieties. In addition to the bile acids, strong signals originated from head (cholin) and glycerophosphoric acid (a variety of fatty acids) groups in phosphatidylcholine were also detected. Chemical shifts information of the phospholipid were summarized in Table 3.

Discrimination analysis between gallstone and gallbladder polyp group

Even though we conducted the metabolic finding with different samples of gallbladder diseases (gallstones and gallbladder polyps), it was hard to find significant spectral and/or metabolic differences between the groups, resulting in the need of multi-variate discrimination analysis, especially the principle component analysis (PCA). In order to remove unrelated spectral variations stemmed from slightly different temperature, pH, and magnetic field drift, the original spectra were reprocessed with binning with 0.006 ppm and aligning along with the chemical shift. Signal intensities were normalized against total integration values so as to eliminate concentration variations derived from the sample preparation steps. With the proper pre-processing, PCA discrimination analysis was performed for the 34 different bile samples. Figure 4 showed the result of discrimination analysis examined with score values of principle components 1 and 2 (PC1 and PC2). The numbers in the parentheses indicate the overall spectral variations corresponding to the each principal component. For clear comparison, data point in each group marked with different colored symbols in the PCA score plot. The majority of the dataset showed reliable score distributions in their particular regions, while some of them were overlapped in left part of the PCA score plot. For thorough examination, it needed to deal with score distributions of the gallstone group because the group was divided into three major areas (left, middle, and right). Because of limited number of samples and types (cholesterol, pigment, and mixed gallstone), it could not be thoroughly investigated in here, however, data points in the left region were mainly originated from the samples of cholesterol gallstone; while pigment or pigment/cholesterol mixed gallstones were positioned in the middle and right regions of score plot. The gallbladder polyps were

Table 3. Chemical shift information of phosphatidylcholine in gallbladder bile.

^{1}H/ppm	δ ^{13}C/ppm	Assignment
0.89	14.3	$\underline{CH_3}$–$(CH_2)_n$
1.29	22.9	CH_3–$\underline{CH_2}$–$(CH_2)_n$
1.59	25.4	$\underline{CH_2}$–CH_2–CO
1.27	27.8, 30.0, 32.2	$(CH_2)_n$
1.60	25.4	CH_2
2.03	27.5	CH_2–$\underline{CH_2}$–CH=CH
2.30, 2.38	34.4	CH_2–$\underline{CH_2}$–CO
3.68	66.3	N–$\overline{CH_2}$
4.03	64.0	CH–$\underline{CH_2}$O
4.25, 4.46	63.2	CH–$\overline{CH_2}$O
4.31	59.7	CH_2–$\underline{CH_2}$–OCH_2
3.24	54.2	N–$\overline{(CH_3)_3}$
5.32	70.8	CHO
5.30, 5.31	128.0, 129.8	$\underline{CH}$=CH–CH_2–CH=$\underline{CH}$
2.74	25.8	$\overline{CH}$=CH–$\underline{CH_2}$–CH=$\overline{CH}$

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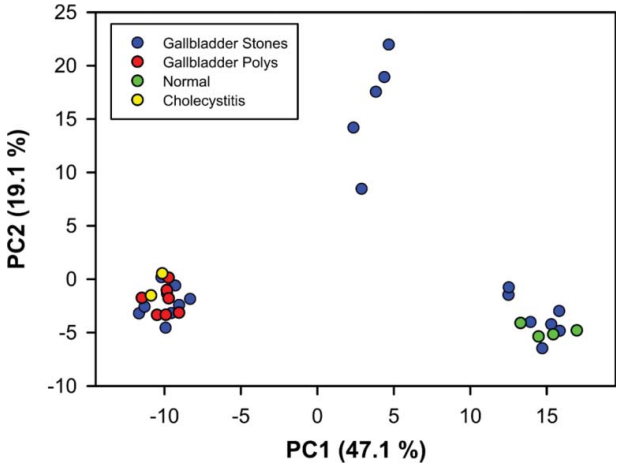

Figure 4. Two-dimensional score plot of PCA discrimination analysis. Each group of gallbladder diseases was displayed with different colored circles.

partially overlapped with gallstone in the left region since three patients in the group had also gallstone, causing the discrimination analysis more complicated and challenging. Therefore, a discrimination accuracy classifying the gallstone and gallbladder polyp groups should be investigated. In the PCA examination, all of the spectra were represented into new extracted features (score values), and then these features were used to classify the samples as either gallstone or gallbladder polyp using SVM. SVM is one of the most common supervised classifiers, in which the best hyper-plane can be effectively determined by maximizing the distance between class boundaries, in order to classify the two groups. The first five principal component scores were fed into SVM, and discrimination accuracy was obtained by four-fold cross-validation with 2,000 runs. A calibration set included 75% of the total samples (2 samples out of the 7 GB polyp samples in total, 16 samples out of the 21 GB stone samples in total), and the remaining samples were used for validation. The discrimination accuracy obtained by the cross-validation was 76.3%.

Moreover, we inspected a PCA loading plot which represents contribution factors for the group clustering as a function of chemical shifts. In Figure 5, loading values of the PC1 and PC2 were showed with absolute positive and negative numbers, respectively, giving straightforward interpretation of the most distinct spectral variables. As highlighted with green colored boxes, four dominant chemical shift regions (PC1; 0.66–0.73 ppm, 0.88–1.08 ppm, 1.29–1.41 ppm, and 3.23–3.28 ppm, PC2; 0.66–1.74 ppm 0.88–1.10 ppm, 1.29–1.41 ppm, and 3.22–3.28 ppm) were identified, which corresponded to aliphatic signals of various cholic acids and phospholipids. Although overall weights of the variable contributions between PC1 and PC2 were relatively different, the active impact regions were almost identical, resulting in importance of the metabolites for data clustering.

In overall, NMR spectroscopy in combination with the metabolic analysis revealed that amphiphilic metabolites in bile, specifically various cholic acids and phospholipids, play a leading role in differentiating gallbladder diseases in molecular levels. Generally, relative concentrations of the amphiphilic surfactants in gallstone samples were slightly lower than the values in gallbladder polyp samples, inducing the crystallization process of fat-like metabolites. Moreover, we found that relative ratios of the amphiphilic metabolites were

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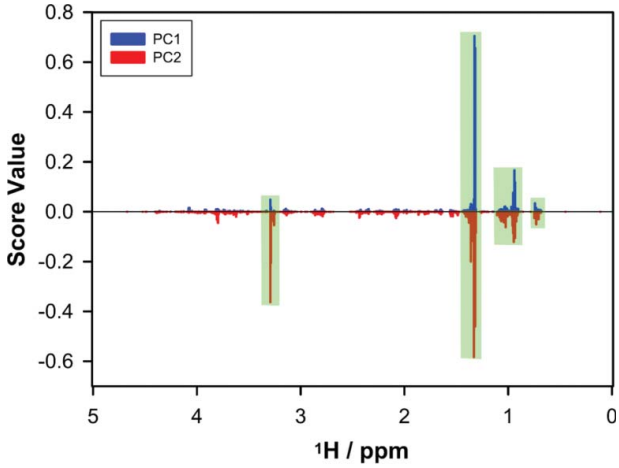

Figure 5. PCA loading plot as a function of chemical shift ranges. Each loading value of the PC1 and PC2 represented with absolute positive and negative numbers, respectively. Main contribution regions for group clustering were highlighted with green colored boxes.

significantly different between different types of gallstones (cholesterol, pigment, and mixed gallstone), which give an explanation of precise information about pathological conditions.

Conclusions

In this study, we performed a metabolic analysis of human gallbladder bile samples using 1D ^{1}H and 2D ^{1}H-^{13}C heteronuclear NMR spectroscopy. Principal component analysis was performed to investigate the spectral variations of different gallbladder diseases. A support vector machine revealed the discrimination accuracy of data clustering. In combination with the discrimination analysis and metabolic profiling, we identified that cholic acids and phospholipids play significant roles in differentiating gallbladder diseases. This information provides potentially valuable information about pathological conditions. Further research is required to elucidate more detailed compositional information with higher discrimination accuracy in order for it to be diagnostically valuable.

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